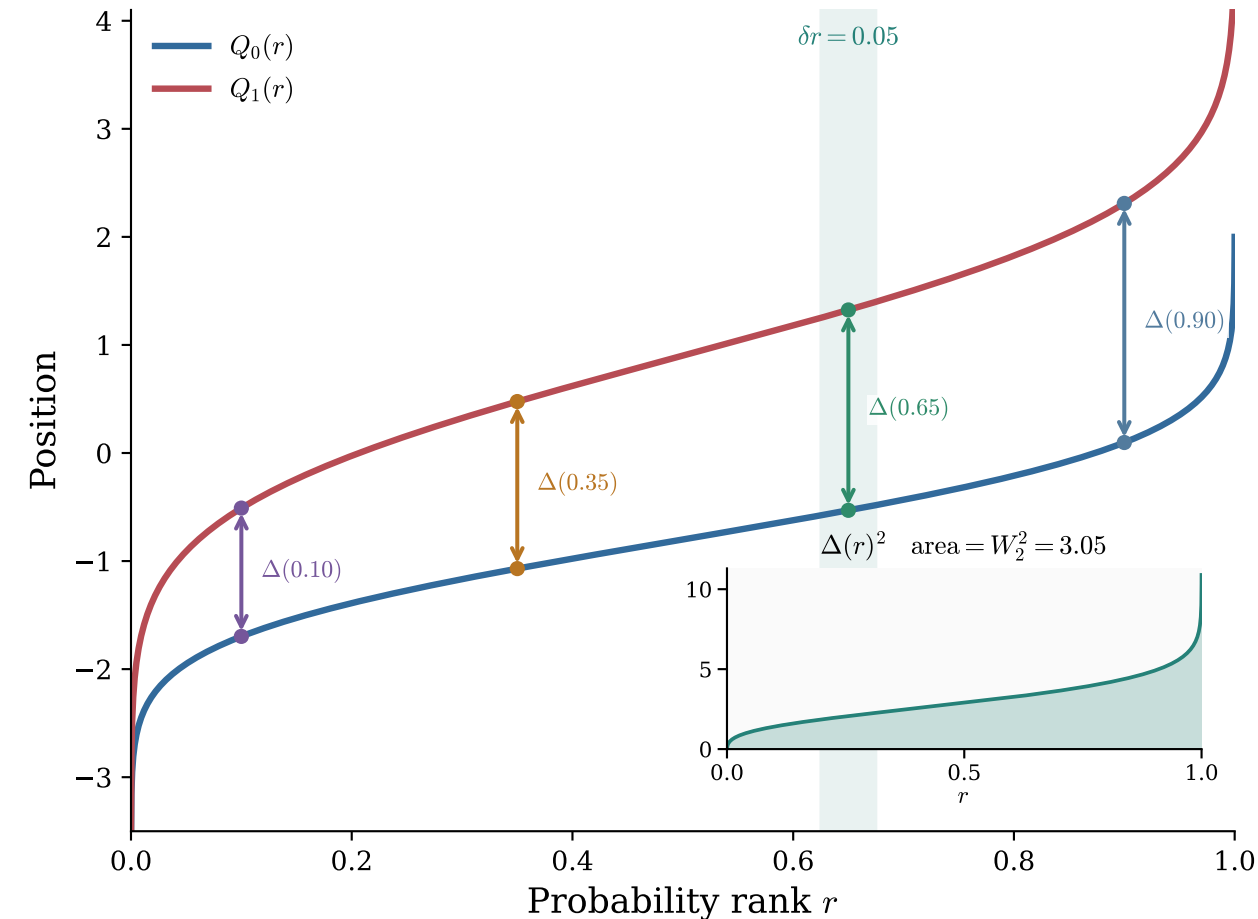


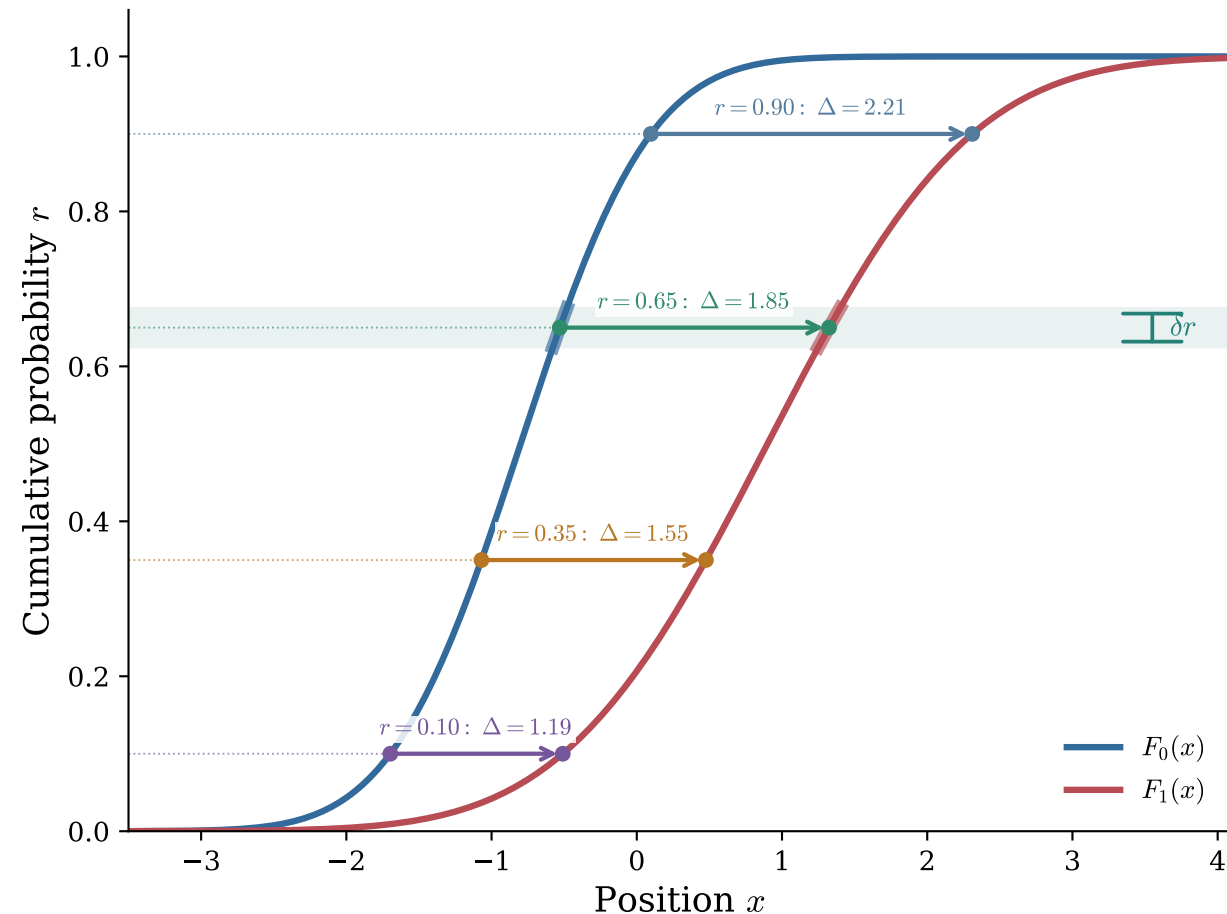
One-dimensional Wasserstein distance: matching equal probability ranks

$$p_0 = \mathcal{N}(-0.8, 0.7^2), \quad p_1 = \mathcal{N}(0.9, 1.1^2)$$

(a) Quantiles: vertical displacements



(b) CDFs: horizontal transport



$$F_0(Q_0(r)) = F_1(Q_1(r)) = r, \quad \Delta(r) = |Q_1(r) - Q_0(r)|$$

$$W_2^2(p_0, p_1) = \int_0^1 \Delta(r)^2 dr = 3.05, \quad W_2(p_0, p_1) = \sqrt{3.05} \simeq 1.7464$$

The highlighted rank interval carries mass δr ; its cost is approximately $\Delta(r)^2 \delta r$.

Quantile tails are cropped for display; the reported distance uses the full interval (0, 1).